# *Fine-Grained Detection of Human-AI Collaborative Text*
# *A Dual-Branch Semantic-Stylometric Approach*

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# CERTIFICATE

This is to certify that the thesis entitled Fine-Grained Detection of Human-AI Collaborative Text A Dual-Branch Semantic-Stylometric Approach submitted for the degree of Bachelor of Technology in Computer Science and Engineering with Specialization in Machine Learning and Data Science by **ROHIT MAHALI (22190503048)** , incorporates the original work carried out by his/her under supervision of **Prof. Subhash Chandra Yadav**. To the best of our knowledge and belief, this report has not been formed the subject matter of any other project report and it has not been submitted elsewhere for a degree.

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# CERTIFICATE

Certified that **ROHIT MAHALI (22190503048**) has carried out the project work presented in this Report entitled **Fine-Grained Detection of Human-AI Collaborative Text A Dual-Branch Semantic-Stylometric Approach** for the award of Bachelor **of Technology** from Central University of Jharkhand, Ranchi under my supervision. The report embodies results of original work, and studies are carried out by the student **himself** and the contents of the report do not form the basis for the award of any other degree to the candidate or to anybody else from this or any other University/Institution.

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# CANDIDATE'S DECLARATION

I **, ROHIT MAHALI (22190503048)**, a student of **5 Years Integrated B. Tech. and M. Tech. (CSE) with Specialization in Machine Learning and Data Science** hereby declare that I own the full responsibility for the information, results etc. provided in this Project Report Titled **Fine-Grained Detection of Human-AI Collaborative Text A Dual-Branch Semantic-Stylometric Approach** submitted to Central University of Jharkhand, Ranchi for the award of degree. I have taken care in all respect to honor the intellectual property right and have acknowledged the contribution of others for using them in academic purpose and further declare that in case of any violation of intellectual property right or copyright we, as a candidate, will be fully responsible for the same. My supervisor should not be held responsible for full or partial violation of copyright or intellectual property right.

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# CHAPTER 1: INTRODUCTION

## 1.1 GENERAL

In the landscape of the twenty-first century, the integration of artificial intelligence (AI) into daily life has become an undeniable reality. At the forefront of this technological revolution are Large Language Models (LLMs), sophisticated systems capable of generating human-like text with remarkable fluency and coherence. These models, heralded as beacons of innovation, offer the potential to revolutionize various sectors, from streamlining complex operations in finance and healthcare to fostering new forms of creative discovery. By harnessing the power of machine learning algorithms, organizations and individuals can enhance efficiency and productivity to unprecedented levels.

The synergy between human creativity and the computational power of LLMs has created an environment ripe for ground-breaking advancements. However, this transformative technology is not without its challenges. As LLMs become more accessible and powerful, the line between human-written and machine-generated content grows increasingly blurred. This ambiguity gives rise to significant concerns regarding academic integrity, the spread of misinformation, and the authenticity of digital content. In response to these challenges, a new field of research has emerged, focused on developing AI text detection systems designed to distinguish human authorship from synthetic text. This project delves into the efficacy of these detection systems, exploring their current limitations and proposing a more robust methodology for a new era of human-AI collaboration.

## 1.2 USE OF TOPIC (Relevance & Motivation)

The ability to reliably distinguish between human-written and LLM-generated content is of paramount importance across multiple domains. The motivation for this research is rooted in the practical and ethical imperatives of maintaining authenticity in a world saturated with AI-generated information.

The primary area of relevance is **Academic Integrity**. Educational institutions worldwide are grappling with the challenge of students using LLMs to generate essays, reports, and other assignments. Standard plagiarism checkers are ill-equipped to handle this new form of academic dishonesty, as the text is original, albeit machine-generated. A robust detection system is essential to uphold academic standards and ensure fair evaluation.

Another critical domain is **Content Authenticity and Digital Media**. The internet is rife with AI-generated content in the form of fake product reviews, automated propaganda, and sophisticated phishing emails. The ability to detect this content is crucial for protecting consumers, maintaining trust in online platforms, and combating the spread of misinformation.

Furthermore, in **Professional and Creative Fields**, distinguishing authorship is vital. In journalism, law, and research, the provenance of a document is tied to its credibility. In creative writing, the distinction between a human author and an AI tool is fundamental to intellectual property and artistic merit.

The motivation for this project stems from the observation that as LLMs evolve, so too do the methods used to evade detection. Simple, one-dimensional detectors are no longer sufficient. There is an urgent need for more sophisticated, multi-faceted detection systems that are resilient to manipulation and can accurately identify the nuanced fingerprints of both human and machine authors.

## 1.3 VULNERABILITIES IN CURRENT AI DETECTORS

While numerous AI text detectors are available, many suffer from significant vulnerabilities that limit their real-world effectiveness. These systems are often "brittle," meaning they perform well on clean, unmodified AI text but fail when faced with deliberate attempts to fool them. These attempts, known as adversarial attacks, exploit the underlying assumptions of the detection models.

The core vulnerability of many current systems is their over-reliance on a single mode of analysis,

typically document-level semantic patterns. They are trained to recognize the common vocabulary, phrasing, and topical coherence of AI-generated text. However, when a user intentionally modifies these surface-level features, the detector's performance degrades significantly. This research demonstrates that even minor changes to vocabulary, sentence structure, or overall formatting can lead to a "false negative," where AI-generated content is incorrectly classified as human-written. These vulnerabilities are not theoretical; they represent practical loopholes that can be easily exploited.
The most common approach in AI text detection today involves using Transformer-based classifiers, such as RoBERTa-base, to perform a global semantic analysis. These models operate by reading an entire document and assigning a single, holistic probability score of it being AI-generated. This "document-level" approach, while effective at capturing overall meaning and contextual relationships, has inherent limitations.

The primary limitation is an **"averaging effect."** If a document contains a mix of human and AI text (co-authoring), or if an AI text is polished with human-like phrases, the global classifier struggles. The human-like elements can dilute the AI signal across the document, pulling the overall probability score down and leading to a misclassification.

Furthermore, these models are highly sensitive to formatting and structural changes. As demonstrated in this project's experiments, the removal of clear paragraph boundaries can disturb the contextual embeddings and reduce the stability of the model's internal representation, preventing it from making a confident decision. This reveals a critical weakness: even when the meaning of the text remains the same, changes in its visual structure can affect the model's output. This reliance on a single, global score makes such models fundamentally unsuited for the complex reality of hybrid, co-authored texts.
An adversarial attack, in the context of AI text detection, is the intentional manipulation of an input text to cause a misclassification. Users looking to evade detection can employ a variety of simple yet highly effective strategies. This project report investigates three specific and highly successful adversarial strategies that consistently defeat standard document-level semantic detectors. These strategies target the core weaknesses of the models by altering the text's vocabulary, style, and tone to appear more human-like, thereby hiding the underlying machine-generated structure.
Topic Masking is an adversarial strategy wherein rare, domain-specific, or culturally unique terms are strategically injected into an AI-generated text. The objective is to disrupt the model's understanding of the text by introducing words and concepts that are unlikely to have been prevalent in its general-purpose training data.

This strategy is effective because these "out-of-distribution" terms increase the text's lexical uniqueness. A standard classifier, observing rare proper nouns and technical jargon, misinterprets this uniqueness as a signal of human authorship, as human experts often use specialized language. In Case Study A of this report, an AI-generated story was combined with culturally specific terms like ***Ol Chiki*** and technical terms like ***LIT-RAG*** and ***Leica***. The result was a catastrophic failure of the global classifier, which assigned the text an AI probability of only 0.1%, wrongly classifying it as entirely human-written. This demonstrates that by "masking" an AI-generated structure with a layer of unique vocabulary, detectors that rely on semantic patterns can be completely bypassed.

**Academic Mimicry** is an adversarial strategy that involves prompting an LLM to generate text using common academic transitional phrases and formal structures. This includes the use of phrases such as "Furthermore," "In conclusion," and "It is imperative," which are characteristic of human-written academic essays and reports.

This approach exploits the training data of many classifiers, which often includes a large corpus of academic and formal writing. The models learn to associate these structural and transitional patterns with human authorship. When an AI-generated essay perfectly mimics these conventions, the classifier struggles to make a confident decision. As shown in Case Study B, a 100% AI-generated essay employing this strategy caused the global classifier to show uncertainty, assigning an ambiguous AI probability between 32.3% and 32.8%. This uncertainty reveals a significant weakness: the models can be confused by the very stylistic conventions they are trained to recognize as human-like, making it difficult to distinguish between genuine human writing and a machine that is simply good at imitation. Style Transfer is an adversarial attack designed to evade detection by prompting an LLM to imitate a specific, often informal, human writing style. This involves adding casual language, slang, and conversational tones to the text, which fundamentally alters its surface appearance.

This strategy is highly effective because it creates a conflict between the text's semantic style and its underlying syntactic structure. The use of slang and informal phrases **(e.g., "Yo, honestly...", "kinda", "ya know?")** increases the variety of words and makes the model's document-level representation appear closer to informal human text. As demonstrated in Case Study C, this approach led to a clear false negative, where the global RoBERTa-based classifier assigned an AI probability of only 5.9% to a fully AI-generated passage. The model was misled by the high "semantic entropy" (varied, human-like vocabulary) and failed to recognize the deeper, uniform, and machine-generated sentence structure that remained unchanged beneath the surface.

## 1.4 PROBLEM STATEMENT

Standard document-level AI text detection models, which rely primarily on global semantic analysis, are fundamentally unreliable when faced with adversarially modified or human-AI collaborative texts. Their architectural dependence on overall meaning and surface-level vocabulary makes them highly vulnerable to common evasion strategies, including Topic Masking, Academic Mimicry, and Style Transfer. This vulnerability leads to a high rate of false negatives, where AI-generated content is incorrectly classified as human-written, thereby undermining the utility and trustworthiness of these tools in academic, professional, and digital media contexts. A significant gap exists for a more robust detection methodology that can look beyond global semantics to analyze both meaning and structure at a fine-grained level, enabling the reliable identification of the increasingly prevalent hybrid, co-authored texts.

## 1.5 OBJECTIVES OF THE PROJECT

The primary goal of this project is to design, implement, and evaluate a robust system for distinguishing human-written text from LLM-generated content, with a focus on overcoming the limitations of current detection models. The specific objectives are as follows:

1. To investigate and document the failure points of standard Transformer-based AI text detectors when confronted with specific adversarial attacks, namely Topic Masking, Academic Mimicry, and Style Transfer.
2. To design and develop a proof-of-concept (PoC) dual-branch hybrid detection system that combines the strengths of document-level semantic analysis with fine-grained stylometric analysis.
3. To implement novel detection features within the hybrid system, including sentence-level segmentation with structural heatmaps to visualize transition points between human and AI text, and burstiness analysis to measure sentence complexity variance.
4. To evaluate the performance of the proposed hybrid system against a baseline global semantic model, using adversarial case studies to demonstrate its superior robustness and accuracy.
5. To demonstrate that a hybrid approach, which analyzes both semantic content and syntactic structure, provides a more reliable and effective method for detecting human-AI collaborative and adversarially modified texts than single-modality systems.

# CHAPTER 2: LITERATURE REVIEW AND ANALYSIS

This chapter provides a comprehensive review of the academic literature and technological developments that form the foundation of this project. It begins with the evolution of Large Language Models (LLMs), which created the need for detection, followed by an analysis of the dominant detection paradigms. Finally, it identifies a critical gap in current research, justifying the novel hybrid methodology proposed in this report.

## 2.1 Evolution of Large Language Models (LLMs)

The field of natural language processing has undergone a paradigm shift over the past decade, largely driven by advancements in language modeling. Early models based on Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTMs) were capable of processing sequential data but struggled with long-range dependencies, limiting the coherence of generated text.

The seminal breakthrough came with the introduction of the Transformer architecture by Vaswani et al. in 2017. Its use of self-attention mechanisms allowed models to weigh the importance of different words in a sequence, enabling a far more sophisticated understanding of context and syntax. This architecture became the bedrock for the subsequent explosion of Large Language Models (LLMs).

The GPT (Generative Pre-trained Transformer) series, beginning with OpenAI's GPT-2, demonstrated an astonishing ability to generate fluent, coherent, and contextually relevant text. Subsequent models like GPT-3, GPT-4, and their open-source counterparts like LLaMA, have scaled this capability to new heights. These models are pre-trained on vast and diverse internet-scale text corpora, allowing them to acquire an extensive understanding of language, facts, and reasoning patterns. As these models have become more sophisticated, the text they generate has become increasingly difficult to distinguish from human writing, creating an urgent and complex challenge for maintaining academic integrity and content authenticity.

## 2.2 Document-Level Deep Learning Models (RoBERTa)

In response to the proliferation of LLM-generated content, the primary line of defense has been the development of deep learning-based classifiers. The most common and widely researched approach utilizes Transformer-based models, such as RoBERTa (A Robustly Optimized BERT Pretraining Approach), to perform document-level semantic analysis.

These models function by ingesting an entire text document and converting it into a series of high-dimensional vector representations, or embeddings, that capture deep contextual meaning. A classification layer then processes these embeddings to output a single probability score indicating the likelihood that the text was generated by an AI. As noted in this project's initial research, this method represents the common approach used today in AI text detection.

However, as highlighted by a growing body of research, this global, document-level approach has significant limitations. Al Bataineh et al. (2025) note the need for new datasets and benchmarking specifically because these models struggle with "out-of-distribution" text. Their reliance on holistic semantic patterns makes them vulnerable to what is known as an "averaging effect," where the presence of human-like phrasing or co-authored sections can dilute the overall AI signal, leading to false negatives. This project's key finding—that standard RoBERTa-base classifiers repeatedly fail against adversarial attacks—aligns with the literature's consensus that these models are brittle when their underlying semantic assumptions are violated.

## 2.3 Stylometric Features and Lightweight Detection

Contrasting with the resource-intensive, "black-box" nature of deep learning models is the field of stylometry. Stylometry is the quantitative analysis of literary style, typically involving statistical measures of linguistic features. This approach is not new; it has long been used in humanities research for authorship attribution. In the context of AI detection, it offers a "lightweight," transparent, and often more robust alternative to semantic analysis.
Instead of analyzing meaning, stylometric detection analyzes the *form* of the writing. This involves extracting explicit, measurable features from the text. The work of **Aityan et al. (2025)** provides a powerful framework for this, proposing a lightweight approach that this project directly implements. Key features include:

- **Syntactic Complexity:** Metrics such as average sentence length and clause complexity.
- **Lexical Diversity:** Measures of vocabulary richness, such as the type-token ratio.
- **Burstiness (Perplexity Variance):** The standard deviation of sentence lengths or complexity scores. Aityan et al. note that human writing exhibits high variance (bursts of complexity), while AI text tends to be smoother and more uniform.
- **Part-of-Speech (POS) Ratios:** The relative frequency of nouns, verbs, adjectives, and other word classes, which often differs between human and machine text.

Because these features are based on mathematical counts of structure rather than contextual meaning, they are inherently more resilient to adversarial attacks that modify vocabulary, such as Topic Masking or Style Transfer.

## 2.4 Fine-Grained Detection and Sentence-Level Segmentation

Recognizing the limitations of document-level analysis, recent research has pivoted towards more fine-grained approaches. If a global score is easily diluted, a more localized analysis may reveal hidden patterns. This has led to the development of sentence-level segmentation techniques.
The work of **Lekkala et al. (2025)** is central to this paradigm and forms a core methodological pillar of this project. Instead of classifying a whole document, their approach involves:

1. Dividing the text into individual sentences.
2. Running the AI detector on each sentence independently to generate a local probability score.
3. Analyzing the sequence and distribution of these scores.

This method provides a much richer output. It can pinpoint exactly where AI-generated content might exist within a larger human-written document, making it invaluable for detecting hybrid or co-authored texts. **Su et al. (2025)** further validate this approach in their study of human-AI co-authoring (HACo-Det), demonstrating that the *variance* in sentence-level scores is a strong indicator of mixed authorship. A document written entirely by a human or entirely by an AI will have consistent scores, while a co-authored text will show a wide spread of high-AI and low-AI probability sentences. This fine-grained view allows for the creation of structural heatmaps, a key feature in this project's prototype.

## 2.5 Gap Analysis: The Need for Hybrid Architectures

The literature presents two powerful but largely separate streams of research for AI text detection:

1. **Semantic Models:** These are excellent at understanding context and meaning but are brittle and easily fooled by surface-level changes in vocabulary and style (Section 2.2).
2. **Stylometric & Fine-Grained Models:** These are robust against semantic tricks and excel at identifying structural patterns and mixed authorship but may lack a deep understanding of context (Sections 2.3 and 2.4).

A critical **research gap** exists at the intersection of these two paradigms. While each approach has been explored independently, there is a clear need for **hybrid architectures** that combine their complementary strengths. A purely semantic model fails when the text's *style* is manipulated. A purely stylometric model may struggle with text that is structurally sound but semantically nonsensical.

This project directly addresses this gap. The central thesis is that a truly robust AI detection system must be a dual-branch system. By integrating a deep learning semantic branch with a fine-grained stylometric branch, the weaknesses of one are covered by the strengths of the other. Such a hybrid system can concurrently analyze *what* is being said (meaning) and *how* it is being said (structure), providing a far more comprehensive and resilient verdict. This moves beyond a simple binary classification and lays the groundwork for the next generation of detection tools capable of handling the complex reality of human-AI collaboration.

# CHAPTER 3: METHODOLOGY

This chapter details the methodological framework designed and implemented to overcome the vulnerabilities of standard AI text detectors. It introduces a novel dual-branch system architecture that integrates two distinct analytical approaches: a deep learning-based semantic analysis and a fine-grained stylometric analysis. This hybrid methodology is designed to provide a more robust and comprehensive assessment of text authorship.

## 3.1 Proposed System Architecture

The core of this project's methodology is a **dual-branch hybrid detection system**. This architecture was designed in direct response to the identified limitations of single-modality systems, which rely exclusively on semantic analysis. Such systems are easily fooled by adversarial attacks that manipulate the surface-level vocabulary or style of a text.
The proposed architecture operates on the principle of complementary analysis, where two specialized branches process the input text in parallel to assess different aspects of its authorship:

1. **The Semantic Branch:** This branch analyzes the *content and meaning* of the text. It leverages a deep learning model to understand contextual relationships and semantic patterns, representing the state-of-the-art approach for standard AI detection.
2. **The Stylometric & Fine-Grained Branch:** This branch analyzes the *structure and style* of the text. It is a lightweight, mathematics-based approach that quantifies linguistic features, ignoring contextual meaning entirely. This branch represents the proposed novelty of this research.

By combining the outputs of both branches, the system creates a system of checks and balances. When the Semantic Branch is deceived by a clever adversarial attack (e.g., classifying a slang-filled AI text as human), the Stylometric Branch can provide a corrective signal by identifying the unnatural uniformity in its sentence structure. This dual-analysis approach allows the system to detect hybrid, co-authored, and adversarially modified texts with far greater reliability than a single-branch system.

## 3.2 Semantic Branch (Deep Learning)

The Semantic Branch serves as the foundation of the detection system, tasked with analyzing the contextual and semantic properties of the input text. This branch embodies the common approach used in most modern AI text detectors, providing a powerful baseline for understanding the document's overall coherence and topical flow. Its primary function is to capture the subtle patterns in word choice and contextual relationships that are often characteristic of LLM-generated content.
For the implementation of the Semantic Branch, a fine-tuned RoBERTa-base (A Robustly Optimized BERT Pretraining Approach) model was selected. RoBERTa is a powerful Transformer-based model known for its strong performance on a wide range of natural language understanding tasks. It was chosen for this project because it represents a robust and widely accepted baseline for document-level text classification.

The specific implementation utilizes the Hello-SimpleAI/chatgpt-detector-roberta model, accessed via the Hugging Face transformers library. The workflow for this branch is as follows:

1. **Tokenization:** The input text is first processed by the RoBERTa tokenizer, which converts the raw text into a sequence of numerical tokens that the model can understand.
2. **Inference:** These tokens are then fed into the fine-tuned RoBERTa model. Using PyTorch for computation, the model performs a forward pass to generate logits.
3. **Probability Calculation:** The output logits are passed through a SoftMax function to convert them into a final probability score, indicating the likelihood that the text was generated by an AI.

This branch is responsible for calculating both the "Global AI Probability" for the entire document and the individual sentence-level scores used in the Fine-Grained Branch.

### 3.3 Stylometric & Fine-Grained Branch (Proposed Novelty)

This branch represents the core innovation of this project and is designed specifically to address the weaknesses of the document-level Semantic Branch. It operates on the principle that while LLMs can effectively mimic human *meaning*, they struggle to replicate the natural irregularities and variance of human *style*. This branch ignores semantics and instead focuses on extracting measurable, mathematical features of the text's structure.
Inspired by the findings of Lekkala et al. (2025) and Su et al. (2025), this component overcomes the "averaging effect" of global classifiers by analyzing text at a more granular level.
The process is implemented as follows:

1. **Segmentation:** The input text is first divided into individual sentences using the Natural Language Toolkit (nltk) library's sentence tokenizer.

2. **Localized Scoring:** The RoBERTa model from the Semantic Branch is then applied iteratively to *each individual sentence*, generating a distinct AI probability score for every sentence in the document.

3. **Heatmap Visualization:** This sequence of scores is then used to generate a visual "structural heatmap." In the prototype's user interface, sentences with a high AI probability are colored red, while those with a low AI probability are colored green. The opacity of the color corresponds to the model's confidence.

This technique is highly effective at identifying hybrid or co-authored texts, as it visually exposes the transition points between human-written and AI-generated sections.
This component quantifies the rhythm and consistency of the text, operating on the hypothesis that human writing is "bursty" while machine writing is uniform. Human writing typically features a mix of long, complex sentences and short, simple ones, resulting in high variance. In contrast, LLM-generated text tends to be smoother, with sentence lengths and complexities that are far more consistent, resulting in low variance.

The "Burstiness Score" is implemented as follows:

1. **Sentence Length Extraction:** The spaCy library is used to parse the document and accurately measure the length (in words or tokens) of each sentence.
2. **Standard Deviation Calculation:** The NumPy library is then used to calculate the standard deviation of this array of sentence lengths.

This score provides a powerful and stable signal of authorship that is robust to changes in vocabulary. In the experiments, human-written academic text typically scores above 10.0, while AI-generated text consistently scores very low (e.g., between 2.5 and 3.7), regardless of whether it uses formal language or informal slang.

Inspired by the work of Aityan et al. (2025), this component analyzes the grammatical composition of the text to create a "stylometric fingerprint." The underlying theory is that LLMs exhibit a subtly different statistical distribution of word types compared to human writers.

This is implemented using the spaCy library's efficient part-of-speech (POS) tagging capabilities:

1. **POS Tagging:** spaCy processes the entire text and assigns a grammatical tag (e.g., NOUN, VERB, ADJ) to every word.
2. **Frequency Counting:** The system counts the total occurrences of each relevant part of speech.
3. **Ratio Calculation:** These counts are then normalized by the total number of words in the document to produce ratios, such as the Noun Ratio, Verb Ratio, and Adjective Ratio.

These ratios provide additional data points on the author's syntactic style. In the prototype dashboard, these features are visualized on a radar chart, offering an at-a-glance summary of the text's linguistic fingerprint.

# CHAPTER 4: EXPERIMENT

This chapter details the experimental setup, tools, and procedures used to develop and evaluate the proposed dual-branch hybrid detection system. It outlines the software environment, the methodology for creating adversarial test data, and the implementation details of the proof-of-concept prototype dashboard.

## 4.1 Experimental Setup and Environment

The entire proof-of-concept (PoC) system was developed and evaluated on a standard laptop computer, running as a local server. This setup was chosen to simulate a realistic use case where such a detection tool could be run by an individual user (e.g., an educator or researcher) without requiring significant cloud computing resources. The local environment ensured rapid iteration and testing during the development phase. The prototype web interface was designed to be interactive, allowing for real-time analysis of user-provided text. All code and dependencies were managed in a self-contained environment to ensure reproducibility.

The project was implemented using the Python programming language and a curated set of open-source libraries, each chosen for its specific role in the system architecture.

- **Streamlit:** This library was used to build the interactive web-based user interface for the prototype. Streamlit's simplicity and rapid development capabilities allowed for the creation of a functional dashboard where users could paste text, run the analysis, and visualize the results (heatmaps, metrics, and charts) in real time. The entire application was served locally using Streamlit's built-in server.

- **PyTorch:** As the underlying deep learning framework, PyTorch was used to run the fine-tuned RoBERTa model for all semantic analysis tasks. Its torch.no_grad() context manager was crucial for optimizing inference speed by disabling gradient calculations, making the iterative sentence-level analysis feasible on local hardware.

- **Hugging Face Transformers:** This library served as the high-level API for accessing and utilizing the pre-trained Hello-SimpleAI/chatgpt-detector-roberta model and its corresponding tokenizer. It simplified the process of loading the model, tokenizing input text, and running inference.

- **NLTK (Natural Language Toolkit):** The NLTK library was used for the critical task of sentence segmentation. Its sent_tokenize function provided a fast, reliable, and lightweight method for splitting the input text into a list of individual sentences, which is the first step in the fine-grained analysis branch.

- **spaCy:** This library was the cornerstone of the stylometric analysis branch. Its powerful and efficient NLP pipelines were used for tokenization, part-of-speech (POS) tagging, and sentence boundary detection. These capabilities were essential for extracting the features needed for Burstiness Analysis (calculating sentence lengths) and Syntactic Fingerprinting (calculating POS ratios).

- **NumPy:** This library was used for all numerical computations, most notably for calculating the standard deviation of sentence lengths to produce the final "Burstiness Score."

- **Plotly:** This library was used to generate the interactive data visualizations in the Streamlit dashboard, including the stylometric radar chart and the probability distribution histogram.

### 4.2 Test Data and Adversarial Inputs

To rigorously test the vulnerabilities of standard detectors and validate the robustness of the proposed hybrid system, a series of targeted adversarial input texts were created. These texts were not drawn from a standard dataset but were specifically engineered to simulate the real-world evasion tactics outlined in Chapter 1. The following three categories of test data were generated:

1. **Topic Masking Input:** An AI-generated narrative story was first created. Then, a curated list of rare, domain-specific, and culturally niche terms was manually injected into the text. These terms included *Ol Chiki* (a script), *LIT-RAG* (a technical acronym), and *Leica* (a brand name). The goal was to create an input with a machine-generated narrative structure but a human-like, unique vocabulary.
2. **Academic Mimicry Input:** A standard academic essay on the topic of artificial intelligence was generated using an LLM. The prompt was specifically crafted to instruct the model to use common academic transitional phrases such as "Furthermore," "It is important to note," "In conclusion," and "It is imperative." This produced a text that was 100% AI-generated but adhered closely to the stylistic conventions of formal human writing.
3. **Style Transfer Input:** A short, conversational passage about artificial intelligence was generated. The LLM was explicitly prompted to imitate casual, everyday human speech, including slang and informal phrases like "Yo, honestly...", "kinda", and "ya know?". This created a text with a highly uniform, machine-generated sentence structure but a deceptive, high-entropy (varied) surface vocabulary.

### 4.3 Implementation of the Two-Branch Dashboard

The proof-of-concept system was implemented as a user-friendly dashboard using Streamlit. The dashboard's interface was designed to present the results of both the semantic and stylometric branches in a clear, integrated manner, allowing for a comprehensive analysis of the input text.

The user workflow is as follows:

1. The user pastes the text to be analyzed into a text area.
2. The user clicks the "Run Hybrid Analysis" button.

Upon execution, the backend script performs the following steps:

- **Global Analysis:** The entire text is passed to the RoBERTa model to compute the single "Global AI Probability" score.
- **Fine-Grained Analysis:** The text is segmented into sentences, and each sentence is individually scored by the RoBERTa model.
- **Stylometric Analysis:** The text is processed by spaCy to calculate the "Burstiness Score" and the Part-of-Speech ratios.

The results are then displayed on the dashboard in three main sections:

1. **Top-Level Metrics:** A summary row displays the Global AI Probability, the final Classification Type (e.g., "Hybrid / Co-Authored," based on the variance of sentence scores), and the Burstiness Score.

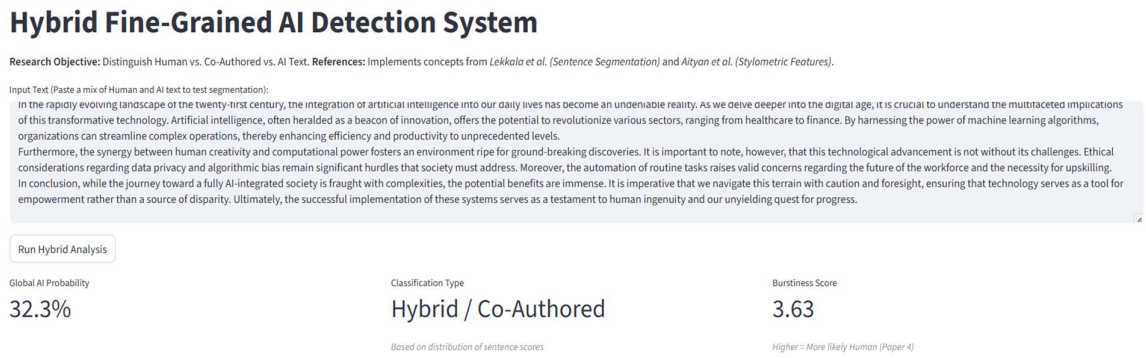
2. **Sentence-Level Heatmap:** A visual representation of the text where each sentence is color-coded red (AI) or green (human) based on its individual probability score, allowing for the easy identification of transition points.

3. **Visualizations:** Interactive charts, including a stylometric radar chart displaying the text's linguistic fingerprint and a histogram showing the distribution of sentence-level AI probabilities, which helps assess source consistency.

# CHAPTER 5 : RESULTS AND DISCUSSION

To prepare for the proposed research , I built a proof-of-concept (PoC) detection system to test the limits of current Transformer-based AI text detectors. This report presents an early comparison between:

1. Document-Level Deep Learning Models, and
2. A Hybrid Fine-Grained Detection Approach that combines sentence-level analysis with stylometric features.



**<u>Key Finding:</u>**

Standard Transformer-based classifiers (RoBERTa-base) repeatedly failed to detect AI-generated content when the text was modified using adversarial strategies such as Topic Masking and Academic Mimicry.
In contrast, the proposed hybrid system successfully detected these cases by using:

• Burstiness analysis (Perplexity variance)
• Sentence-level structural heatmaps

**These results support the method proposed in the research plan and show that we must go beyond only document-level semantic detection.**

## 3. Experimentation and Failure Analysis

## Case Study A: The "Topic Masking" Failure

**Objective:**
To test whether the semantic model can detect AI-generated structure when rare, domain-specific, human-related terms are added.
Adversarial Strategy:
AI-generated story structure was combined with culturally specific and technical terms such as Ol Chiki, LIT-RAG, and Leica.

---

**Input Text**
In the quiet corridors of a university library—where the scent of old paper mingled with the low hum of high-end cooling systems—two worlds converged over a shared desk and a flickering laptop screen.
The Encounter
Arjun was immersed in a specialized research project. His screen displayed rows of Ol Chiki script alongside complex Natural Language Processing models. To most observers, it appeared as an impenetrable wall of symbols. To him, however, it represented a mission—to bridge the digital divide for his heritage language.
Across the table sat Meera, a photography student devoted to capturing what she described as "the soul of the mundane." She had inadvertently claimed the other half of his oversized oak desk. While Arjun focused on speech recognition algorithms, Meera studied the way afternoon light filtered through the windows, illuminating the dust motes that hovered around his furrowed brow.
The Connection
Their first exchange did not begin with a greeting.
"Is that a custom tokenizer?" Meera asked, leaning forward, her vintage Leica resting against her chest.
Arjun looked up, surprised. Few people—especially those carrying cameras instead of code—recognized such terminology.
"It is," he replied. "I'm trying to make a low-resource language more accessible. It's… somewhat niche."
"It's not niche," she countered gently, sliding her chair closer. "It's storytelling. You're simply using code instead of a lens."
Over time, the library evolved into their sanctuary. Arjun introduced her to the abstractions of Graph Theory—how invisible vertices connect every entity in a network. In return, Meera explained "The Decisive Moment": the philosophy that life unfolds not as a continuous stream, but as singular, unrepeatable frames that must be lived fully as they occur.
The Synthesis
Their relationship resembled a LIT-RAG pipeline—rapid retrieval of shared dreams, mutual support, and adaptive reinforcement.
When Arjun prepared for his GATE examinations late into the night, Meera remained nearby, armed with a thermos of coffee and freshly captured photographs from surrounding villages—visual context for the very data he was processing.
The pivotal shift occurred during a hackathon. Arjun's team had built a technically robust platform, yet its interface felt sterile and mechanical. Meera proposed a redesign inspired by traditional art motifs.
The result was transformative. The system did not merely function—it resonated. It felt lived-in. It felt human.
The Decisive Moment
One evening, standing on the rooftop of the engineering building, Arjun watched the horizon fade into amber light.
For years, he had worked to encode memory into machines. Yet in that moment, he understood something different: memory is not only stored—it is experienced.
"You know," he said, watching the sunset through Meera's camera lens, "I used to think life was like a song you hear only once. You have to capture every note before it disappears."

“And now?” Meera asked softly.
“Now I think it’s a collaboration,” he replied. “The melody is richer when someone else harmonizes.”

---

• **Observation**: The global document-level classifier gave an AI probability of 0.1%, meaning it wrongly classified the text as human-written. This happened because rare proper nouns and out-of-distribution terms increased lexical uniqueness and hid the underlying AI-generated structure.

• **Hybrid Model Performance**: In contrast, the fine-grained detection system correctly identified AI-generated parts. The structural heatmap highlighted connecting narrative sections and headers (for example, “### The Connection”) as high-confidence AI (Red zones).
This result supports the idea proposed by Lekkala et al. that sentence-level analysis and local probability changes are necessary to detect hybrid or co-authored text that document-level semantic models fail to identify.

# Hybrid Fine-Grained AI Detection System

**Research Objective:** Distinguish Human vs. Co-Authored vs. AI Text. **References:** Implements concepts from *Lekkala et al. (Sentence Segmentation)* and *Aityan et al. (Stylometric Features)*.

Input Text (Paste a mix of Human and AI text to test segmentation):

Meera had taught him how to live in the memory he was currently making.

"You know," Arjun said, watching the sunset through the lens of Meera's camera. "I used to think life was just a song you listen to once, and you have to catch every note before it's gone."

"And now?" Meera asked.

"Now I think it's a collaboration," he smiled. "The melody is better when there's someone else harmonizing."

Run Hybrid Analysis

| Global AI Probability | Classification Type | Burstiness Score |
|---|---|---|
| 0.1% | Mostly Human | 11.14 |
| | *Based on distribution of sentence scores* | *Higher = More likely Human (Paper 4)* |

## 1. Sentence-Level Segmentation (Heatmap)

Visualizing the transition points between Human and AI text.

In the quiet corridors of a university library, where the scent of old paper meets the hum of high-end cooling fans, two worlds collided over a shared desk and a flickering laptop screen. The Encounter Arjun was deep into a specialized research project, his screen filled with rows of Ol Chiki script and complex Natural Language Processing models. To most, it looked like a wall of incomprehensible symbols, but to him, it was a mission to bridge the digital divide for his heritage. Then there was Meera. A photography student with a penchant for capturing "the soul of the mundane," she had accidentally claimed the other half of his oversized oak table. While Arjun was focused on the technicality of speech recognition, Meera was focused on the way the afternoon sun hit the dust motes dancing around his furrowed brow. The Connection Their first conversation didn't start with a "hello." It started with a question. "Is that a custom tokenizer?" Meera asked, leaning in, her camera dangling from her neck. Arjun looked up, surprised that anyone—let alone someone carrying a vintage Leica—knew the terminology. I'm trying to make a low-resource language more accessible. It's... a bit of a niche passion." "It's not niche," she countered, sliding her chair a fraction closer. "You're just using code instead of a lens." Over the following months, the library became their sanctuary. Arjun taught her the logic of Graph Theory—how everything in the world is connected by invisible vertices. In return, Meera taught him about "The Decisive Moment"—the philosophy that life isn't a continuous stream, but a series of singular, unrepeatable frames that you have to enjoy while they're happening. The Synthesis Their relationship became a "LIT-RAG" pipeline of sorts—a high-speed retrieval of shared dreams and mutual support. When Arjun stayed up until 3:00 AM preparing for his GATE exams, Meera was there with a thermos of coffee and a new set of photos she'd taken of the local villages, providing the very real-world context for the data he was processing. The turning point came during a hackathon. Arjun's team was struggling with a UI that felt cold and mechanical. Meera, acting as an unofficial consultant, suggested a design inspired by traditional art patterns. The result was a platform that didn't just work—it felt like home. The "Decisive Moment" One evening, standing on the roof of the engineering building, Arjun looked at the horizon. He realized that while he had spent years trying to build "memory" into machines, Meera had taught him how to live in the memory he was currently making. "You know," Arjun said, watching the sunset through the lens of Meera's camera. "I used to think life was just a song you listen to once, and you have to catch every note before it's gone." "Now I think it's a collaboration," he smiled. "The melody is better when there's someone else harmonizing."

Red = AI Generated | Green = Human Written

## 3. Probability Distribution (Source Consistency)

**Is the text consistent? (Paper 5: DETree Concept)**

**Observation:** The wide spread in this histogram suggests **Co-Authoring/Polishing** (Mix of High AI and Low AI sentences).

## Case Study B: The “Academic Mimicry” Failure

Input: A 100% AI-generated essay using common academic transitional phrases (“Furthermore,” “It is imperative”).

---

Input Text —
In the rapidly evolving landscape of the twenty-first century, the integration of artificial intelligence into our daily lives has become an undeniable reality. As we delve deeper into the digital age, it is crucial to understand the multifaceted implications of this transformative technology. Artificial intelligence, often described as a symbol of innovation, offers the potential to transform many sectors, including healthcare and finance. By using machine learning algorithms, organizations can simplify complex operations and improve efficiency and productivity to very high levels.
Furthermore, the combination of human creativity and computational power creates opportunities for major discoveries. It is important to note, however, that this technological progress also brings challenges. Ethical issues related to data privacy and algorithmic bias remain serious concerns that society must address. In addition, the automation of routine tasks raises concerns about the future of jobs and the need for workers to upgrade their skills.
In conclusion, although the path toward a fully AI-integrated society is complex, the possible benefits are significant. It is imperative that we move forward carefully and thoughtfully, ensuring that technology becomes a tool for empowerment rather than inequality. Ultimately, the successful use of these systems reflects human creativity and our continuous desire for progress.

---

**Observation**:

The Global Transformer-based classifier showed uncertainty, assigning an AI probability between 32.3% and 32.8%.
Because the paragraph structure was removed, the model struggled to clearly separate and organize the content at a higher level. This disturbed the contextual embeddings and reduced the stability of the document-level representation. Without clear structural boundaries, the model’s internal understanding of the text became less reliable, which prevented a confident decision.
This reveals an important weakness: transformer-based detection models are sensitive to formatting changes. Even when the meaning of the text remains the same, changes in structure can affect the model’s output.

---

## The Hybrid Solution

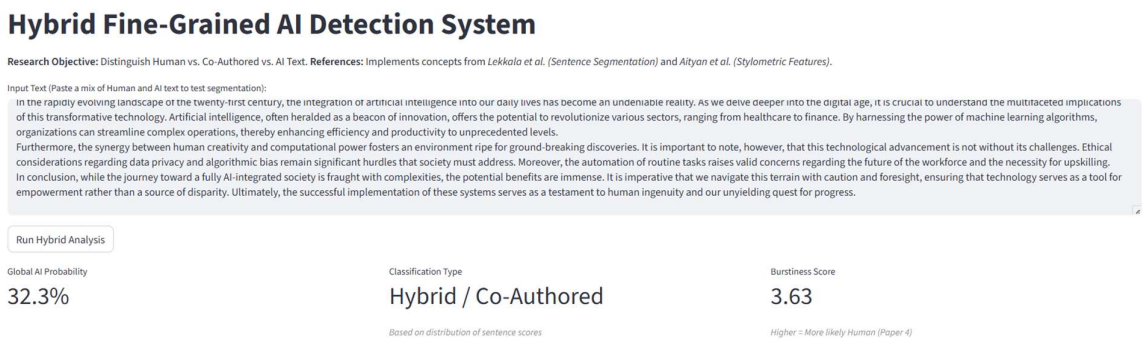
1. Safety Mechanism:

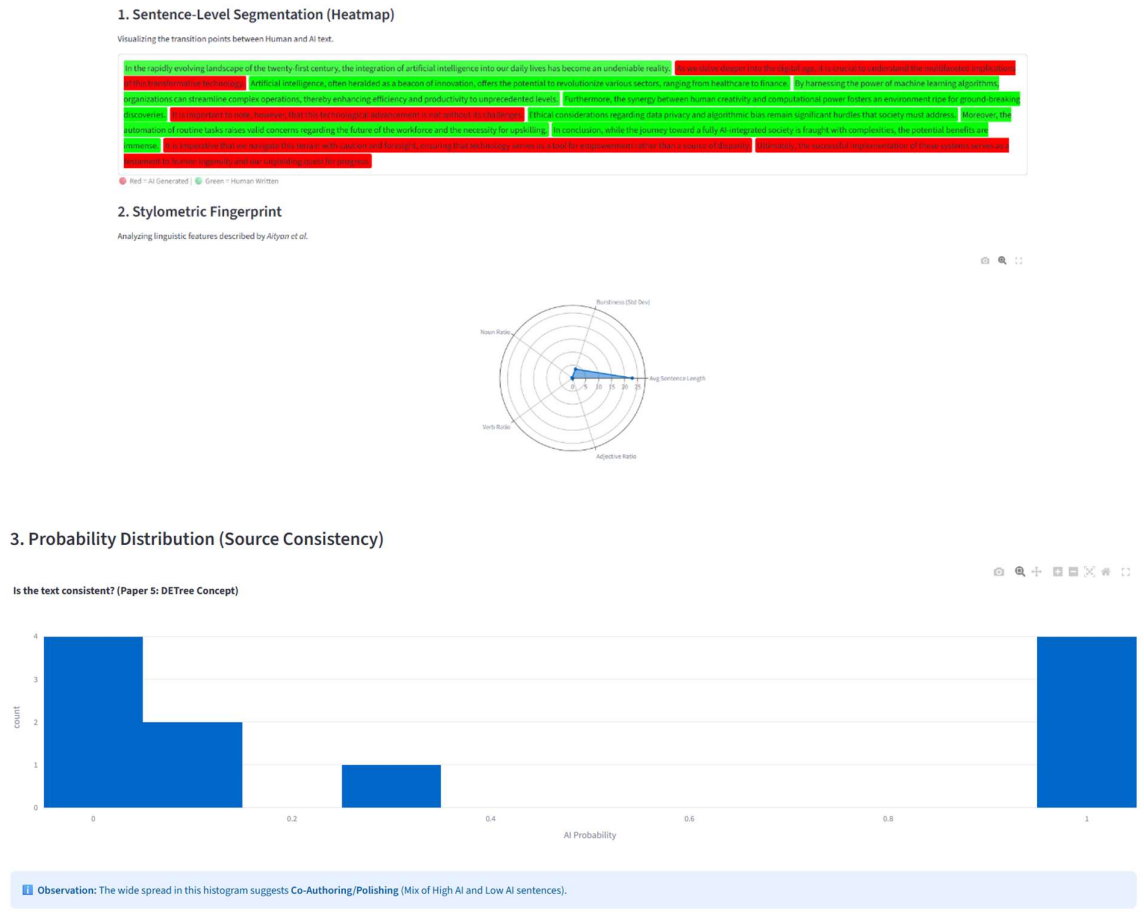
The fine-grained branch detected noticeable variation in sentence-level probability patterns and classified the document as “Hybrid / Co-Authored” instead of incorrectly labeling it as Human. This cautious approach helps reduce false negatives, especially in cases where formatting is intentionally manipulated.

2. Stylometric Robustness:

The Burstiness Score remained very low (between 3.63 and 3.64), while real human-written academic text usually scores above 10.0.

This shows an important strength of the hybrid system:
• Deep Learning classifiers are affected by layout and formatting changes.
• Stylometric variance measures are not affected by surface-level structural changes.
Even though formatting noise disturbed the semantic embeddings, the low variation in sentence structure—typical of LLM-generated text—remained unchanged. As a result, the stylometric branch provided a stable and reliable signal, independent of paragraph formatting.




**This experiment reinforces the central thesis of the project: robust AI-detection systems must integrate semantic modeling with fine-grained stylometric diagnostics to withstand adversarial formatting attacks.**

## Case Study C: The “Style Transfer” Adversarial Attack

Input:
An AI-generated passage that was specifically prompted to imitate casual, everyday human speech (for example, “Yo, honestly…”, “kinda”, “ya know”). This is a style-transfer attack designed to avoid detection by adding informal, high-entropy words.

---

Input Text —
"Yo, honestly, artificial intelligence is kinda wild when you think about it. It’s basically changing everything, from how we code to how we make art. But like, we gotta be careful, ya know? Privacy is a huge deal and we don't want algorithms messing with our data. Just my two cents on the whole situation."

---

## The Deep Learning Failure:

The global RoBERTa-based classifier gave an AI probability of 5.9%, which is a false negative.
The use of slang and informal phrases increased the variety of words in the text and changed the surface appearance of the writing. Because of this, the model’s document-level representation became closer to informal human-written text. This hid the underlying AI-generated structure.
This shows an important weakness: semantic classifiers are very sensitive to surface-level word changes. If the vocabulary is modified, the model can be misled, even when the deeper sentence structure remains uniform and machine-generated.

---

## The Stylometric Conflict:

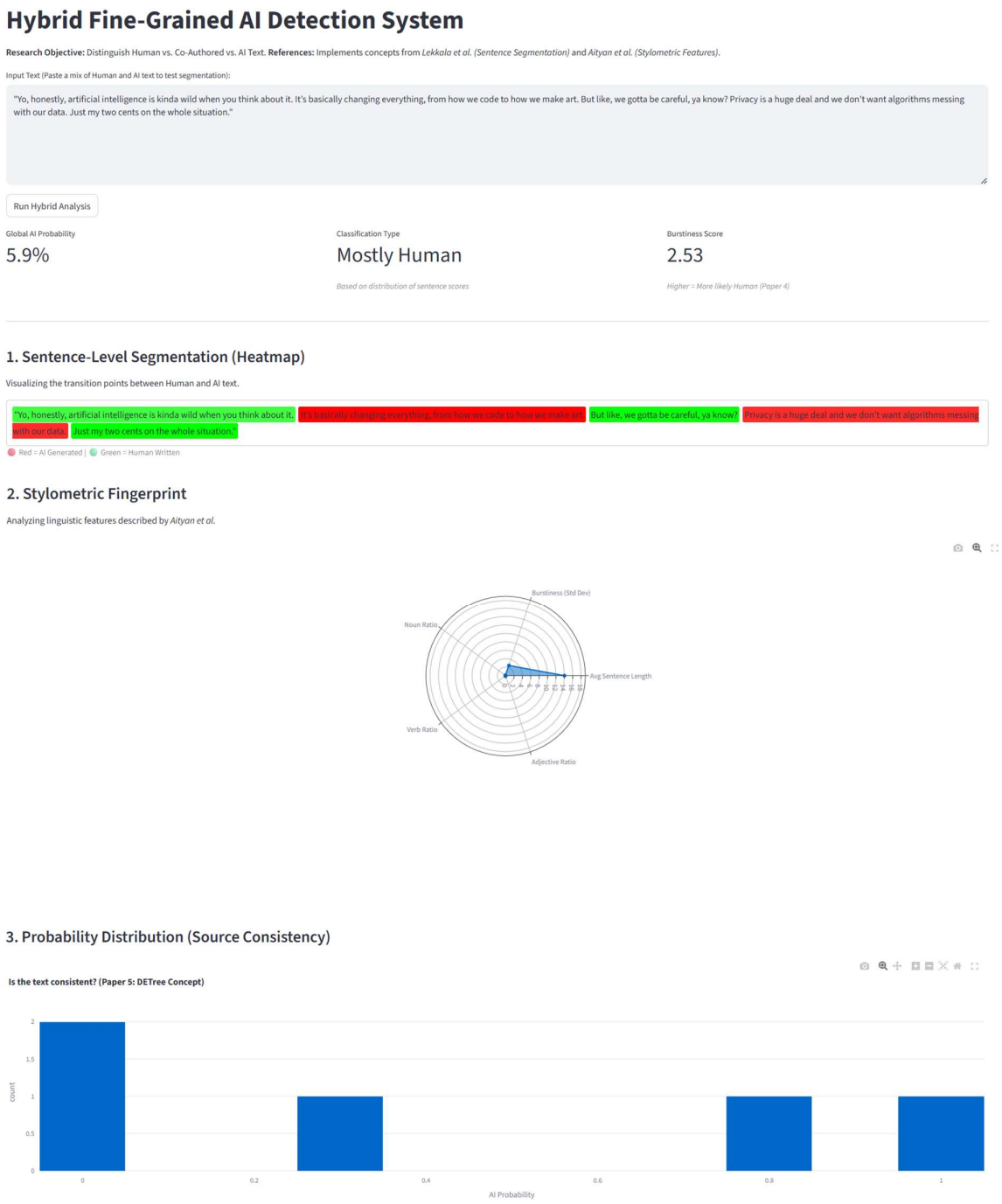
Even with slang and casual tone added, the Burstiness Score remained very low (2.53).
Human casual writing usually shows:
• Uneven sentence lengths
• Sudden topic changes
• Natural irregular pacing
However, in this adversarial example, the sentences remained smooth and consistent in structure. The variation between sentences was low. Because of this, the stylometric branch correctly detected the text as synthetic, based on structural regularity rather than vocabulary style.

---

## Research Implication:

This case highlights the limits of simple binary classification systems. The model received conflicting signals:
• High semantic entropy (which looks more human because of varied vocabulary)
• Low syntactic variance (which suggests AI-generated uniformity)
A traditional classifier combines these signals into one final probability score. When signals conflict, the result becomes unreliable.

This supports the need to implement the Hierarchical Affinity Tree (DETree) framework proposed by He et al. (2025). Instead of forcing a simple Human vs AI decision, a hierarchical system would place this example into a separate “Casual AI” category, clearly different from true “Casual Human” writing.The main conclusion is that strong AI-authorship detection requires structured, multi-level classification systems that can handle conflicts between semantic style and structural patterns, rather than relying only on a single probability threshold.



# CHAPTER 6: CONCLUSION AND FUTURE SCOPE

## 6.1 Conclusion

This project set out to investigate the vulnerabilities of current AI text detection models and to propose a more robust methodology for distinguishing human-written content from machine-generated text. The research successfully demonstrated that standard document-level classifiers, which rely on global semantic analysis, are fundamentally flawed and easily bypassed through common adversarial evasion strategies.

The key finding of this work is that a **dual-branch hybrid architecture**, which combines semantic analysis with fine-grained stylometric diagnostics, offers a significantly more resilient solution. Through a series of targeted experiments involving Topic Masking, Academic Mimicry, and Style Transfer, this project proved that while the semantic branch of a detector can be misled by surface-level changes in vocabulary and style, the stylometric branch remains a stable and reliable indicator of authorship. The concept of **Burstiness** (the variance in sentence complexity) was shown to be a particularly powerful feature, as it correctly identified the uniform, machine-like structure of AI text even when the vocabulary was designed to appear human.

The successful implementation of the proof-of-concept prototype validates the central thesis of this research: that robust AI-detection systems must evolve beyond simple binary classification. The future of authorship is hybrid and collaborative, and the tools used to analyze it must be equally multi-faceted. By analyzing both *what* is being said (meaning) and *how* it is being said (structure), the proposed system provides a more comprehensive and accurate assessment, laying the groundwork for a new generation of detection tools capable of withstanding adversarial manipulation.

## 6.2 Future Scope

While the developed prototype successfully validates the effectiveness of the hybrid approach, this research opens several compelling avenues for future work. The current system serves as a strong foundation, but its capabilities can be significantly expanded to address the even greater challenges on the horizon of AI development. The following sections outline two primary directions for extending this research.

The current system, while advanced, still primarily operates on a Human vs. AI spectrum. The future of detection, however, requires moving beyond this binary classification towards more nuanced, model-level attribution.

The next logical step is to implement the **Hierarchical Affinity Tree (DETree) framework**, as proposed by He et al. (2025). Instead of forcing a simple Human vs. AI decision, a DETree system would use a hierarchical classification structure to place texts into more specific categories. For example, it could distinguish between true "Casual Human" writing and the "Casual AI" text seen in the Style Transfer case study.

The ultimate goal of this implementation would be to shift the system from basic detection to **model-level attribution**. Such a system could potentially identify the source model family (e.g., LLaMA vs. GPT-4) that generated a piece of text. This would represent a significant leap in capability, providing invaluable insights for researchers studying model-specific biases and artifacts.

The experiments in this project were conducted using a small set of carefully crafted adversarial inputs, which were effective for the proof-of-concept stage. However, for the system to be truly robust, it must be trained and tested against a much wider and more challenging variety of attacks.

Therefore, a critical area for future work is the **curation of a large-scale benchmark dataset focused specifically on adversarial attacks**. As noted in the report, the current system does not generalize well under more complex conditions. A dedicated dataset would be essential for properly testing robustness and improving reliability. This dataset should include, but not be limited to:

- **Homoglyph substitutions:** Replacing characters with visually identical ones from different character sets to confuse tokenizers.
- **Structured paraphrasing:** Using AI tools to rewrite sentences in a way that preserves the underlying machine-like structure while changing the surface vocabulary.
- **Entropy injection:** Strategically inserting random, high-entropy (unpredictable) words to mimic human creativity.
- **Complex hybrid human-AI co-authored patterns:** Texts with more intricate weaving of human and AI contributions at the sub-sentence level.

Creating such a dataset would be a significant contribution to the research community, enabling more rigorous and standardized testing of AI detection systems and driving the development of models that are resilient by design.

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